HOMEWORK 6 FOR MATH 2860 (ELEMENTARY DIFFERENTIAL EQUATIONS)

E. A. ATINDAMA, PHD MATHEMATICS

Due at 10:00 EST in class

Question Q1. Given $y_1(x) = x^2$ is a solution of the ODE

$$x^2y'' - 3xy' + 4y = 0,$$

(a) Find a second solution to the ODE.
(b) What is the domain/interval on which the two solutions form a fundamental set of solutions? (In other words, on what domain/interval are the two solutions linearly independent?)
(c) Find the general solution of the ODE on the interval from part (b).

Show all your work for full credit.

Solution
(a) First, we use reduction of order to find the other solution.
Set $y_2 = v(x)y_1 = v(x)x^2$, substitute and reduce order to find $v'(x) \propto x^{-3}$;
Integrating yields $v(x) = \ln x$, which implies that $y_2 = x^2 \ln x$.
(b) $y_1 = x^2$ and $y_2 = x^2 \ln x$. Compute the Wronskian to determine linear independence.

$$y_1' = 2x, \qquad y_2' = 2x \ln x + x.$$

The Wronskian is

$$W(y_1, y_2)(x) = \begin{vmatrix} x^2 & x^2 \ln x \\ 2x & 2x \ln x + x \end{vmatrix} = x^2(2x \ln x + x) - (x^2 \ln x)(2x) = x^3.$$

Since $W(x) = x^3 \neq 0$ for every $x \neq 0$, the two solutions are linearly independent on any interval that does not contain 0.
Thus they form a fundamental set on, for example, $(0, \infty)$. You may choose $(-\infty, 0)$ if you represent $y_2 = x^2 \ln |x|$.
(c) The general solution on such an interval (take $(0, \infty)$) is

$$\boxed{y(x) = C_1x^2 + C_2x^2 \ln x, \qquad x > 0,}$$

and similarly on $(-\infty, 0)$ (with $\ln |x|$ if preferred).

Question Q2. Is $y_p = \frac{11}{12}x + \frac{1}{2}x$ a particular solution of the nonhomogeneous equation $y''' - 6y'' - 11y' + 6y = -3x$?

Show all your work for full credit.

(E. A. Atindama) . Email addresses: emmanuel.atindama@utoledo.edu
Version 0.0.1. Last updated: September 28, 2025.

Solution
Given the candidate particular solution:

$$y_p(x) = \tfrac{11}{12} + \tfrac{1}{2}x, \quad y_p' = \tfrac{1}{2}$$

Higher derivatives of $y_p = 0$. Thus, substituting into the LHS yields

$$-11\left(\tfrac{1}{2}\right) + 6 \cdot \left(\tfrac{11}{12} + \tfrac{1}{2}x\right) = -\tfrac{11}{2} + \tfrac{66}{12} + \tfrac{6}{2}x = 3x$$

Hence the proposed $y_p = \frac{11}{12} + \frac{1}{2}x$ is *not* a particular solution.

Question Q3.
(a) Show that $2\cos x = e^{ix} + e^{-ix}$
(b) Show that $\dfrac{d}{dx}(\sin x) = \dfrac{e^{ix} + e^{-ix}}{2}$
Show all your work for full credit.

Solution
(a) Euler's formulas give

$$e^{ix} = \cos x + i\sin x, \qquad e^{-ix} = \cos x - i\sin x.$$

Adding these two identities cancels the sine terms and doubles the cosine:

$$e^{ix} + e^{-ix} = 2\cos x \quad \Longrightarrow \quad \boxed{2\cos x = e^{ix} + e^{-ix}.}$$

(b) Using the exponential representation of $\sin x$,

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}.$$

Differentiate term-by-term:

$$\frac{d}{dx}\sin x = \frac{d}{dx}\left(\frac{e^{ix} - e^{-ix}}{2i}\right) = \frac{ie^{ix} - (-i)e^{-ix}}{2i} = \frac{ie^{ix} + ie^{-ix}}{2i} = \frac{e^{ix} + e^{-ix}}{2}.$$

Using part (a) the right-hand side equals $\cos x$, so this recovers the familiar identity $(\sin x)' = \cos x$. Thus

$$\boxed{\frac{d}{dx}\sin x = \frac{e^{ix} + e^{-ix}}{2}.}$$